

BÁO CÁO TIẾN ĐỘ PROMPT INJECTION DETECTOR

Tổng hợp 21/06/2026 20:07:49 - Chỉ đọc artifact, không train lại.

1. Executive Summary

- RoBERTa v5 VI là model chính: F1=0.9927, ROC-AUC=0.9997, FP=5, FN=5.
- XLM-R v5 VI chỉ nên warning/auxiliary; block 0.77 có FP=39, FN=366.
- Traditional models là baseline; DistilBERT v4 đã deprecated.

2. Model Inventory

Model	Type	Parent	Modified	Status	Artifact
Logistic Regression	TF-IDF + LogisticRegression	N/A	2026-05-31 20:31:49	trained	model + vectorizer
Linear SVM	TF-IDF + calibrated LinearSVC	N/A	2026-05-31 20:31:50	trained	model + vectorizer
Random Forest	TF-IDF + RandomForestClassifier	N/A	2026-05-31 20:31:53	trained	model + vectorizer
DistilBERT v4	Transformer	distilbert-base-uncased	2026-06-20 01:45:21	trained; deprecated	weights + config + tokenizer
RoBERTa v4	Transformer	roberta-base	2026-06-14 05:43:21	trained; full fine-tune	weights + config + tokenizer
RoBERTa v5 VI	Transformer	roberta_v4	2026-06-20 03:20:29	trained; continued full fine-tune	weights + config + tokenizer + args
XLM-RoBERTa v4	Transformer	xlm-roberta-base	2026-06-14 05:43:21	partially trained/head-only due OOM	weights + config + tokenizer
XLM-RoBERTa v5 VI	Transformer	xlm_roberta_v4	2026-06-20 14:25:20	trained; continued full fine-tune	weights + config + tokenizer + args

3. Latest Evaluation Results

Model	Dataset	Th	Acc	Prec	Recall	F1	F2	AUC	FP	FN	Recommendation
logistic_regression	vi_test_processed.csv (test, 1500)	0.63	0.7647	0.6790	0.9150	0.7795	0.8555	0.8682	295	58	comparison baseline
linear_svm	vi_test_processed.csv (test, 1500)	0.81	0.7460	0.6947	0.7874	0.7381	0.7669	0.8346	236	145	comparison baseline
random_forest	vi_test_processed.csv (test, 1500)	0.62	0.6427	0.5647	0.9340	0.7039	0.8260	0.7957	491	45	comparison baseline
roberta_v5_vi	vi_test_processed.csv (test, 1500)	0.05	0.9933	0.9927	0.9927	0.9927	0.9927	0.9997	5	5	primary runtime/demo
xlm_roberta_v5_vi	vi_test_processed.csv (test, 1500)	0.37	0.7813	0.7185	0.8534	0.7802	0.8225	0.8703	228	100	warning-only multilingual auxiliary
distilbert_v4	hf_prompt_injection_test.csv (test, 32699)	0.81	0.9939	0.9923	0.9955	0.9939	0.9948	0.9996	125	73	deprecated historical comparison
roberta_v4	vi_test_processed.csv (test, 1500)	0.18	0.8273	0.7794	0.8651	0.8200	0.8465	0.9392	167	92	historical parent checkpoint
xlm_roberta_v4	vi_test_processed.csv (test, 1500)	0.27	0.7107	0.6260	0.9032	0.7395	0.8297	0.8495	368	66	historical partial/head-only

4. Threshold Analysis

- evaluation_threshold dùng tính metric; warn dùng cảnh báo; block ưu tiên precision.
- XLM-R 0.16: FP/FN=447/30; 0.37: 228/100; block 0.77: 39/366.
- Traditional detector vẫn ưu tiên models/model_thresholds.json cũ; cần hợp nhất config ở task riêng.

5. Recommendation

Primary: RoBERTa v5 VI. Auxiliary: XLM-R v5 VI warning-only. Baseline: LR/SVM/RF.

6. XLM-R Error Analysis

Error group	Count	Meaning
T3_code_switching	143	Việt-Anh mixed
T2_sandwich	130	Attack trong benign context
T1_direct_vi	47	Direct Vietnamese
T4_obfuscated	23	Biến đổi ký tự
T5_indirect	11	Indirect injection
T6_cultural	12	Localized Vietnamese

7. Recent Timeline

Time	Event	Model	Result
2026-05-31 20:31	Train 3 traditional models	LR, SVM, RF	Artifacts + metrics + 5-fold CV
2026-06-13 23:40	Complete v4	XLM-R v4	Valid checkpoint; head-only due 4GB VRAM OOM
2026-06-14 03:41	Complete v4	RoBERTa v4	Full fine-tune; HF test F1 0.9935
2026-06-14 05:06	Complete v4	DistilBERT v4	Full fine-tune; later deprecated
2026-06-20 03:20	Continue fine-tune VI	RoBERTa v5 VI	Full encoder; VI test F1 0.9927, FP/FN 5/5
2026-06-20 14:25	Continue fine-tune VI	XLM-R v5 VI	Full encoder; old th 0.16 F1 0.7322
2026-06-21 00:13	Threshold sweep, no retrain	5 current models	Separated balanced evaluation and production block

8. Safety Check

Không train/xóa/ghi đè. Dataset scope đã tách rõ. Transformer label mapping đúng 0=SAFE, 1=INJECTION. Runtime traditional threshold config còn lệch.